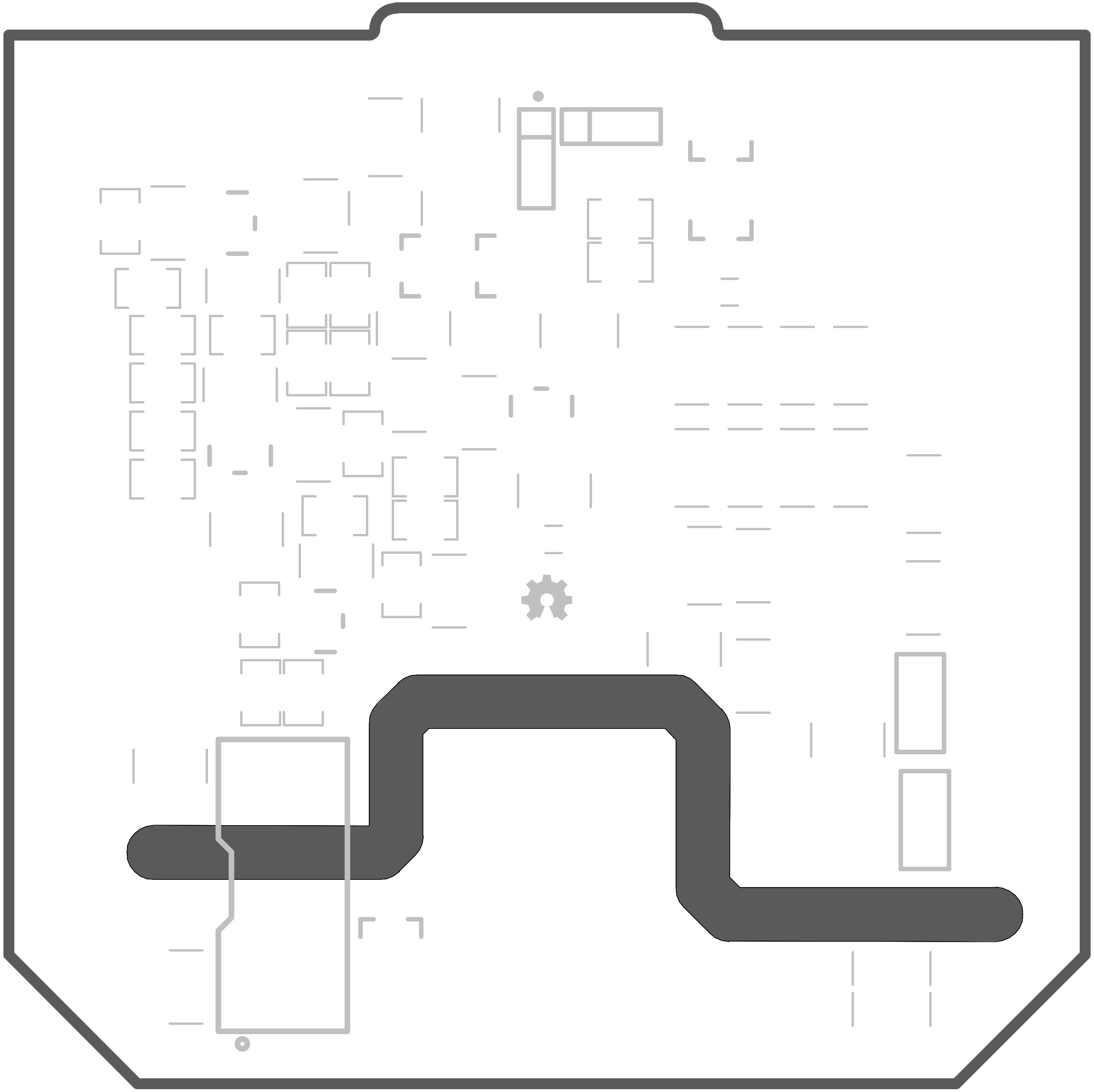
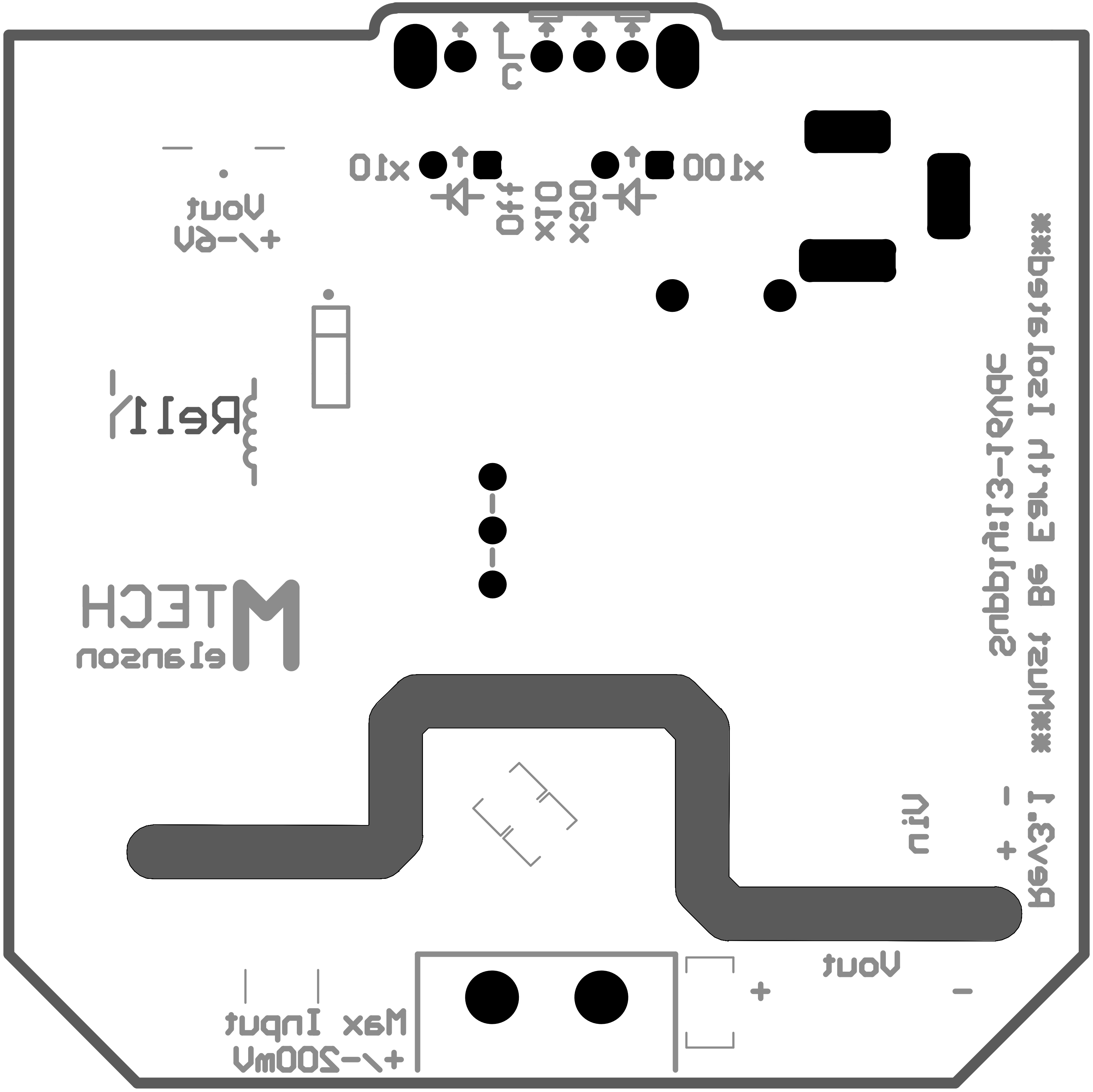
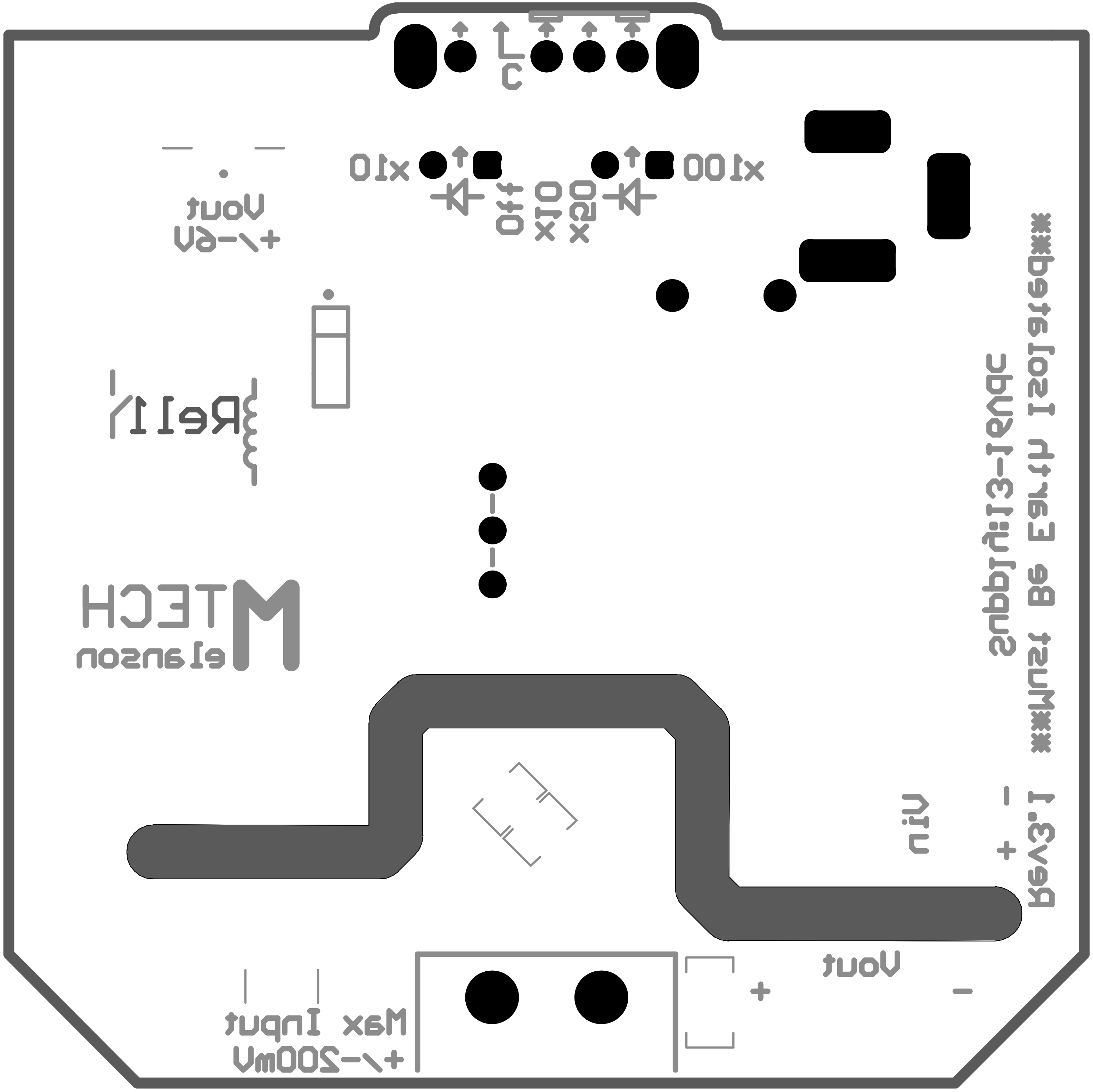
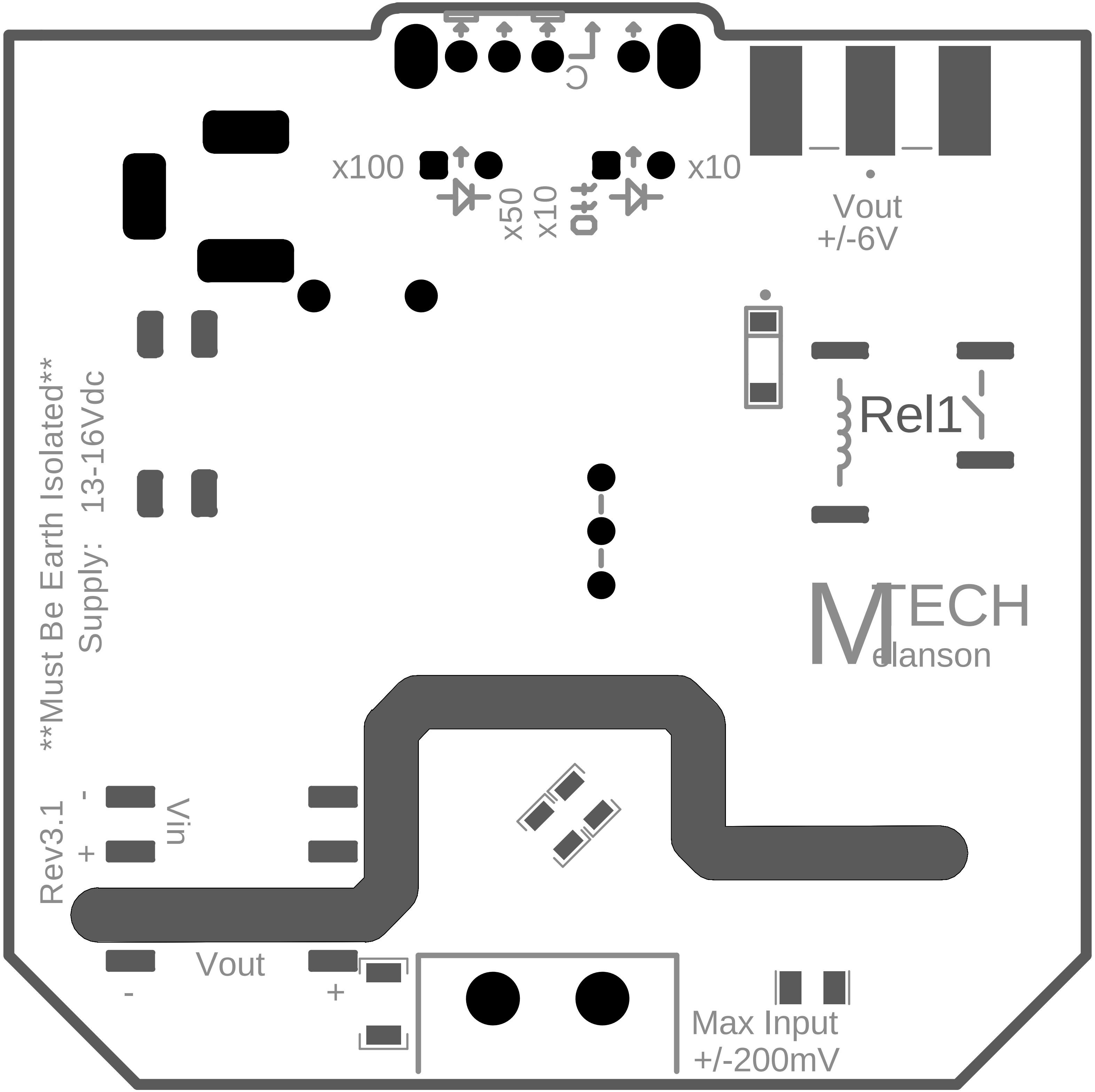



L
C
x10
x100
Off
x10
x50
Vout
+/-6V
Rel1
MTECH
elanson
Rev3.1 **Must Be Earth Isolated**
Vin
+
-
Vout
+
-
Max Input
+/-200mV


L
C
x10
x100
Off
x10
x50
Vout
+/-6V
Rel1
MTECH
elanson
Rev3.1
Must Be Earth Isolated
Supply:3-lead
Vin
+
-
Vout
+
-
Max Input
+/-200mV


x100
x10
x50
x10
Vout
+/-6V
Rel1
MTECH
Melanson
Must Be Earth Isolated
Supply: 13-16Vdc
Rev3.1
Vin
-
+
Vout
-
+
Max Input
+/-200mV

Bill of Materials

<Parameter Title not found>

Source Data From: MelansonTech Isolated Current Sense Rev 3.PrjPcb
Project: MelansonTech Isolated Current Sense Rev 3.PrjPcb
Variant: None

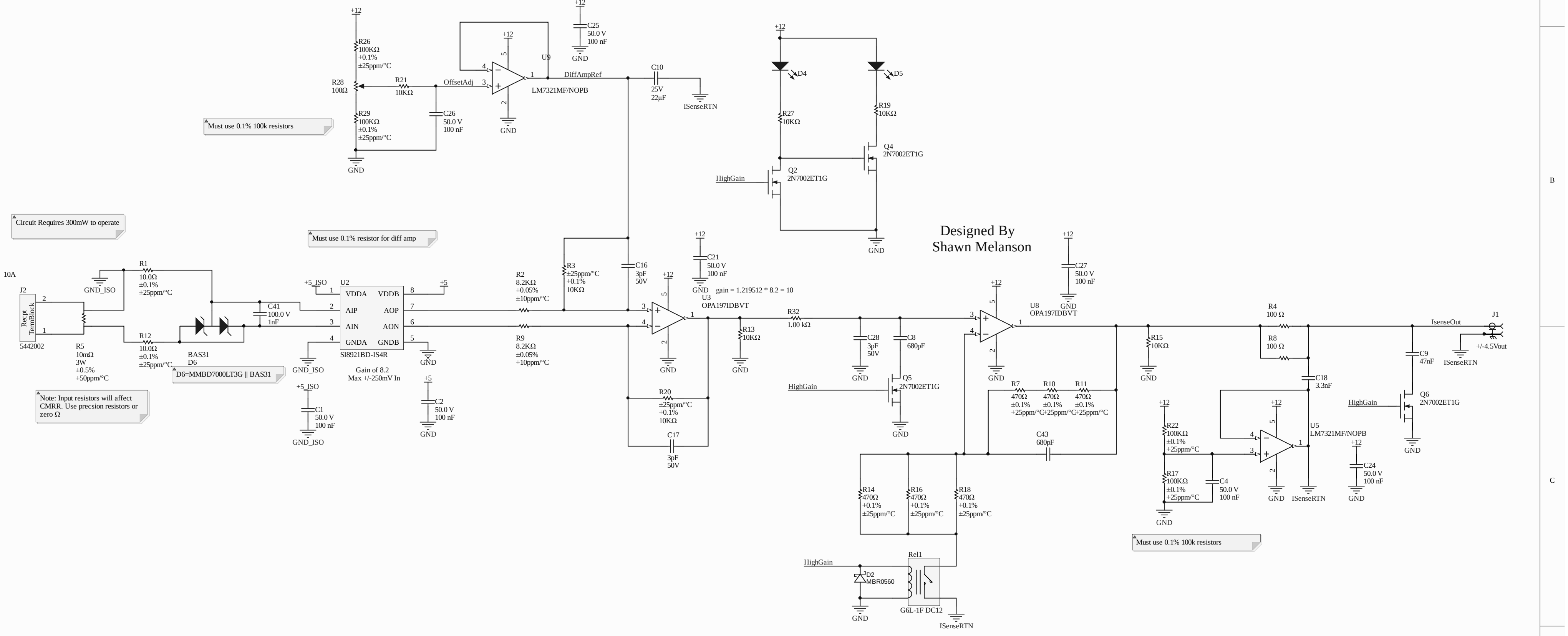
Creation Date: 2026-08-20 8:10:27 PM
Print Date: 20-Aug-26 8:10:51 PM

Production Quantity: 1
Currency CAD

Footprint	Comment	LibRef	Designator	JLCPCB Part #	Description	Quantity	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Order Qty 1	Supplier Subtotal 1	Manufacturer 1	Manufacturer Part Number 1
CAP, 0805, HD	CC0805KRX7R9BB104	Cap,0805,100nf,50V,5%,CC0805KRX7R9BB104	C1, C2, C4, C5, C6, C19, C21, C24, C25, C26, C27, C31, C32		CAP CER 0.1UF 50V X7R 0805	13	Digikey	311-1140-6-ND	0.10554	13	$ 1.37	Yageo Group	CC0805KRX7R9BB104
Cap,0805	CL21A226MAQNNNE	Cap,0805,22uF,25V,X5R,±20%,CL21A226MAQNNNE	C3, C14, C22, C29, C30, C33, C34, C35, C36, C37, C38, C39, C40		CAP CER 22UF 25V X5R 0805	13	Mouser	187-CL31B226KPHNNNE				Loading...	Loading...
Cap,5mm Lead Spaceing,14mm HIGH,10MM Dia, Polymer - RA1	A750MV108M1EAAE014	Cap,Polar,TH,1000uF,25V,Polymer,±20%,A750MV108M1EAAE014	C7		1000 µF 25 V Aluminum - Polymer Capacitors Radial, Can 14mOhm 2000 Hrs @ 105°C	1	Digi-Key	399-A750MV108M1EAAE014-ND	1.80524	1	$ 1.81	Yageo Group	A750MV108M1EAAE014
CAP, 0805, HD	Cap	CAP,0805,Generic	C8, C43			2							
CAP, 0805, HD	CL21B473KCFNNNE	Cap,0805,47nF,100V,±10%,CL21B473KCFNNNE	C9	C53134	CAP CER 0.047UF 100V X7R 0805	1	Digikey	1276-2960-1-ND	0.13886	1	$ 0.14	Samsung Electro-Mechanics	CL21B473KCFNNNE
Cap,0805	CL21A226MAYNNNE	Cap,0805,22uF,25V,X5R,±20%,CL21A226MQQNNNE	C10	C45783	CAP CER 22UF 25V X5R 0805	1	DigiKey	1276-CL21A226MAYNNNECT-ND	0.36105	1	$ 0.36	Samsung Electro-Mechanics	CL21A226MAYNNNE
Cap, 1206	CL31B475KBHNNNE	Cap,1206,4.7uF,10%,50V,CL31B475KBHNNNE	C11, C12	C29823	Cap,SMD,MLCC,4u7,50V,X7R,10%,1206	2	LCSC	C51205				Supplier disabled	Supplier disabled
Cap,0805	CC0805KRX7R9BB104	Cap,0805,100nf,50V,5%,CC0805KRX7R9BB104	C13		CAP CER 0.1UF 50V X7R 0805	1	Digikey	311-1140-6-ND	0.19441	1	$ 0.19	Yageo Group	CC0805KRX7R9BB104
CAP, 0805, HD	CC0805CRNPO9BN3R0	CAP,0805,3pF,COG,NPO,CC0805CRNPO9BN3R0	C16, C17, C28	C519937	CAP CER 3PF 50V C0G/NPO 0805	3	Future Electronics	9724337				Loading...	Loading...
CAP, 0805, HD	CC0805KPX7R9BB332	Cap,0805,3.3nF,50V,±10%,CC0805KPX7R9BB332	C18	C1738	CAP CER 3300PF 50V X7R 0805	1	Digikey	13-CC0805KPX7R9BB332CT-ND	0.18052	1	$ 0.18	Yageo Group	CC0805KPX7R9BB332
CAP, 0805, HD	C0805X102J1GACAUTO	Cap,0805,1nF,±5%,100Vdc,CL10F104ZB8NNNC	C41	C46653	CAPACITOR, 1NF, 100V, NP0, 0805	1	Digi-Key	399-19430-1-ND	0.97205	1	$ 0.97	Yageo Group	C0805X102J1GACAUTO
CM Choke,SMD,DR331	DR331-105BE	common mode, 1mH@100kHz, 500mA, SMD,2ln, DR331-105BE	CM1		CMC 1MH 500MA 2LN SMD	1							
SOD123	MBR0560	Diode,SOD123,Shottky,60V,0.5A,MBR0560	D1, D2, D3	C77338	Dio,Sig,Schtky,0.5A,60V,SOD123	3	Digi-Key	MBR0560TPMSCT-ND	0.27773	3	$ 0.83	MCC	MBR0560-TP
LED,TH,T1,3MM-Vert	OVLBB4C7	LED,TH,T1,3mm,Blue,OVLBB4C7	D4, D5		Blue 470nm LED Indication - Discrete 3.2V Radial	2							
SOT23-3	BAS31	Diode,Standard,Array,Series,120V,200mA,BAS31	D6	C415998	Diode Array 1 Pair Series Connection 120 V 200mA Surface Mount TO-236-3, SC-59, SOT-23-3	1	DigiKey	BAS31FSDKR-ND	0.63878	1	$ 0.64	onsemi	BAS31
CINCH_142-0701-801	142-0701-801	Conn,SMA,Female,End Launch,142-0701-801	J1		CONN SMA JACK STR 50OHM EDGE MNT	1	Digi-Key	J502-ND	8.66518	1	$ 8.67	Cinch Connectivity Solutions	142-0701-801
TERM BLOCK HDR 2POS 5.08MM,5442002	5442002	TermBlock,Header,Male,RA,TH,12A,400V,5.08mmPitch,5442002	J2		TERM BLOCK HDR 2POS 5.08MM	1	Allied	71064909		100		Phoenix Contact	5442002
Connector,Barrel,2.1mm,5.5mm,DC power Jack	PJ-002A	Connector,Barrel,ID 2.1mm,OD 5.5mm,Power Connect, PJ-002A	J4		Power Barrel Connector Jack 2.10mm ID (0.083"), 5.50mm OD (0.217") Through Hole, Right Angle	1	Digi-Key	CP-002A-ND	0.86096	1	$ 0.86	Same Sky	PJ-002A
Ind, 0805	MLZ2012M2R2HT000	Ind. 0805,2.2uH,600mA,160mOhm,MLZ2012M2R2HT000	L1		FIXED IND 2.2UH 600MA 160 MOHM	1	Digi-Key	445-8658-1-ND	0.13886	1	$ 0.14	TDK	MLZ2012M2R2HT000
Ind, 0805	MLZ2012N6R8LT000	Ind,0805,6.8uH,550mA,0.25O,MLZ2012N6R8LT000	L2, L3		FIXED IND 6.8UH 550MA 250 MOHM	2	Digi-Key	445-6761-1-ND	0.13886	2	$ 0.28	TDK	MLZ2012N6R8LT000
Symbol,Open Source - No Text	Open Source Logo	Symbol,Open Source	Logo1		Open Source Silk Screen Logo	1							
14-SMD Module, 6 Leads,NXJ1S0505T6MC-R7	NXJ1S0505T6MC-R7	Mod,PSU,Isolated,4.5-5Vin,5Vout,1W,4.2kVIso,200mA,NXJ1S0505T6MC-R7	PSU1		Isolated Module DC DC Converter 1 Output 5V 200mA 4.5V - 5.5V Input	1	DigiKey	811-NXJ1S0505T6MC-R7CT-ND	3.7077	1	$ 3.71	Murata Power Solutions	NXJ1S0505T6MC-R7
SOT23-3	2N7002ET1G	MOSFET,N-Channel,SOT23,0.25A,60V,2N7002ET1G	Q2, Q4, Q5, Q6	C71533	Small Signal MOSFET, 60 V, 310 mA, Single, N-Channel, 3-Pin SOT-23, Pb-Free, Tape and Reel	4	Digi-Key	2N7002ET1GOSCT-ND	0.31939	4	$ 1.28	onsemi	2N7002ET1G
res, 0805, HD	RNCS0805BKE10R0	Res,0805,10.0O,0.1%,±25ppm/°C,0.1W,RNCS0805BKE10R0	R1, R12	C346661	RES 10 OHM 0.1% 1/10W 0805	2							
res, 0805, HD	ERA-6VRW8201V	Res,0805,8.2KO,±0.05%,±10ppm/°C,ERA-6VRW8201V	R2, R9	C543754	8.2 kOhms ±0.05% 0.25W, 1/4W Chip Resistor 0805 (2012 Metric) Anti-Sulfur, Automotive AEC-Q200 Thin Film	2	DigiKey	10-ERA-6VRW8201VCT-ND	0.84708	2	$ 1.69	Panasonic	ERA-6VRW8201V
res, 0805, HD	RG2012P-103-B-T5	Res,0805,10.0kO,0.1%,±25ppm/°C,RG2012P-103-B-T5	R3, R20	C328719	RES SMD 10K OHM 0.1% 1/8W 0805	2	Future Electronics	4000243				Loading...	Loading...
res, 0805, HD	CRCW0805100RJNEAC	Res,0805,100O,5%,±200ppm/°C	R4, R8	C17408	RES 100 OHM 5% 1/8W 0805	2	Digi-Key	541-100ACT-ND	0.13886	2	$ 0.28	Vishay	CRCW0805100RJNEA
Res,SMD,3637,Nonstandard,4Term	WSL3637R0100DEA	Res,3637,10mO,3W,±0.5%,±50ppm/°C,WSL3637R0100DEA	R5	C4188216	10 mOhms ±0.5% 3W Chip Resistor Nonstandard Anti-Sulfur, Automotive AEC-Q200, Current Sense, Moisture Resistant, Pulse Withstanding Metal Element	1							
Res,1206	RMCF1206FT220R	Res,1206,220O,±1%,±100ppm/°C,0.25W,RMCF1206FT220R	R6	C2484105	220 Ohms ±1% 0.25W, 1/4W Chip Resistor 1206 (3216 Metric) Automotive AEC-Q200 Thick Film	1	DigiKey	RMCF1206FT220RCT-ND	0.13886	1	$ 0.14	Stackpole Electronics	RMCF1206FT220R
res, 0805, HD	ERA-6AEB471V	Res,0805,470O,±0.1%,±25ppm/°C,ERA-6AEB471V	R7, R10, R11, R14, R16, R18	C473280	RES SMD 470 OHM 0.1% 1/8W 0805	6							
res, 0805, HD	CRCW080510K0FKTA	Res,0805,10.0kO,1%,100ppm,CRCW080510K0FKTA	R13, R15, R19, R21, R27	C17414	RES SMD 10K OHM 1% 1/8W 0805	5	Digi-Key	541-3016-1-ND	0.11665	10	$ 1.17	Vishay	CRCW080510K0FKTA
res, 0805, HD	ERA-6AEB104V	Res,0805,100K,±0.1%,±25ppm/°C,ERA-6AEB104V	R17, R22, R26, R29	C192074	RES SMD 100K OHM 0.1% 1/8W 0805	4	Mouser	667-ERA-6AEB104V				Loading...	Loading...
res, 0805, HD	Res1	Res,0805,Generic	R23			1							
Res,TH,POT,TopAdj	CT94EW103	Res,POT,TH,10KO,0.5W,±10%,±100ppm/°C,CT94EW103	R28		TRIMMER 10K OHM 0.5W PC PIN TOP	1	Digi-Key	CT94EW103-ND				Loading...	Loading...
res, 0805, HD	Vishay CRCW08051K00JNEA	Res,0805,1.00kO,5%,200ppm,CRCW08051K00JNEA	R32	C17513	Res Thick Film 0805 1K Ohm 5% 0.125W(1/8W) ±200ppm/C Molded SMD Automotive Paper T/R	1	Digi-Key	541-1.0KADKR-ND	0.13886	1	$ 0.14	Vishay	CRCW08051K00JNEA
Relay,SMD,G6L-1F 12DC	G6L-1F DC12	Relay,SPST,1A,SMD,60Vdc,12Vcoil,Non Latching,G6L-1F DC12	Rel1		Telecom Relay SPST-NO (1 Form A) Surface Mount	1	Allied	70175325		1		Omron Electronic Components	G6L-1F DC12
Switch,Slide,.SP3T, OS103011 -Side	OS103011MS8QP1	Switch,TH,Slide,SP3T,100MA,12V,OS103011MS8QP1	SW1		SWITCH SLIDE SP3T 100MA 12V	1	Digi-Key	CKN9543-ND	1.30533	1	$ 1.31	C&K Components	OS103011MS8QP1
SOT89-3	UA78L12ACPKG3	Reg,Linear,Pos,Fixed,12Vout,27Vin,100mA,UA78L12ACPKG3	U1	C7372179	IC REG LINEAR 12V 100MA SOT89-3	1	Digi-Key	296-50454-6-ND	0.52769	1	$ 0.53	Texas Instruments	UA78L12ACPKG3
SOIC-8 DWV	SI8921BD-IS4R	IC,CurrentSense,Isolated,8.2 Gain,500Khz,SI8921BD-IS4R	U2		Isolation IC Inverters, Power Supplies, UPS 8-SOIC	1	Digikey	863-SI8921BD-IS4RCT-ND	8.7485	1	$ 8.75	Skyworks Solutions	SI8921BD-IS4R
SOT23-5	OPA197IDBVT	OPAMP, single,General Purpose,Rail-to-Rail,10MHz,SOT-23-5,OPA197IDBVT	U3, U8	C131929	General Purpose Amplifier 1 Circuit Rail-to-Rail SOT-23-5	2	LCSC	C131929				Supplier disabled	Supplier disabled
SOT23-5	LM7321MF/NOPB	OPAMP,20Mhz,MOSFET DRIVER,SOT23-5,LM7321MF	U5, U9	C128614	Single Rail-to-Rail Input/Output, ±15V, High Output Current and Unlimited Capacitive Load Operational Amplifier, 5-pin SOT-23, Pb-Free	2	Digi-Key	LM7321MF/NOPBCT-ND				Loading...	Loading...
SOT89-3	HT7550-1	Reg,Linear,Fixed,SOT-89-3,30Vin,5Vout,150mA,HT7550-1	U6	C16106	LOW DROPOUT REGULATORS(LDO) POSITIVE FIXED 55MV @ 1MA 30V 5V 150MA SOT-89(SOT-89-3) ROHS	1							
						95					$ 35.42		

Approved **Notes**

Isolated Shunt Amplifier


+12
R26
100KΩ
±0.1%
±25ppm/°C
R28
100Ω
R21
10KΩ
OffsetAdj
R29
100KΩ
±0.1%
±25ppm/°C
C26
50.0 V
100 nF
GND
+12
U9
LM7321MF/NOPB
DiffAmpRef
+12
C25
50.0 V
100 nF
GND
C10
25V
22µF
ISenseRTN
+12
D4
D5
R27
10KΩ
R19
10KΩ
Q4
2N7002ET1G
Q2
2N7002ET1G
HighGain
GND
10A
J2
Recpt TermBlock
5442002
R1
10.0Ω
±0.1%
±25ppm/°C
GND_ISO
R5
10mΩ
3W
±0.5%
±50ppm/°C
R12
10.0Ω
±0.1%
±25ppm/°C
BAS31
D6
C41
100.0 V
1nF
+5_ISO
U2
VDDA
AIP
AIN
GNDA
VDDB
AOP
AON
GNDB
SI8921BD-IS4R
Gain of 8.2
Max +/-250mV In
GND_ISO
+5
GND
+5_ISO
C1
50.0 V
100 nF
GND_ISO
+5
C2
50.0 V
100 nF
GND
R2
8.2KΩ
±0.05%
±10ppm/°C
R9
8.2KΩ
±0.05%
±10ppm/°C
R3
±25ppm/°C
±0.1%
10KΩ
C16
3pF
50V
+12
GND
R20
±25ppm/°C
±0.1%
10KΩ
C17
3pF
50V
+12
C21
50.0 V
100 nF
GND
gain = 1.219512 * 8.2 = 10
U3
OPA197IDBVT
R13
10KΩ
GND
R32
1.00 kΩ
C28
3pF
50V
GND
C8
680pF
Q5
2N7002ET1G
HighGain
GND
Designed By
Shawn Melanson
+12
C27
50.0 V
100 nF
GND
+12
U8
OPA197IDBVT
GND
R7
470Ω
±0.1%
±25ppm/°C
R10
470Ω
±0.1%
±25ppm/°C
R11
470Ω
±0.1%
±25ppm/°C
C43
680pF
R14
470Ω
±0.1%
±25ppm/°C
R16
470Ω
±0.1%
±25ppm/°C
R18
470Ω
±0.1%
±25ppm/°C
Rel1
HighGain
D2
MBR0560
GND
G6L-1F DC12
ISenseRTN
R15
10KΩ
GND
R4
100 Ω
R8
100 Ω
C18
3.3nF
+12
R22
100KΩ
±0.1%
±25ppm/°C
R17
100KΩ
±0.1%
±25ppm/°C
C4
50.0 V
100 nF
GND
+12
U5
LM7321MF/NOPB
GND
ISenseRTN
+12
C24
50.0 V
100 nF
GND
IsenseOut
J1
+/-4.5Vout
ISenseRTN
C9
47nF
Q6
2N7002ET1G
HighGain
GND

Must use 0.1% 100k resistors

Circuit Requires 300mW to operate

Must use 0.1% resistor for diff amp

D6=MMBD7000LT3G || BAS31

Note: Input resistors will affect CMRR. Use precsion resistors or zero Ω

Designed By
Shawn Melanson

Must use 0.1% 100k resistors

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Title

Size	Number	Revision
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Date: 2026-08-20 Sheet of
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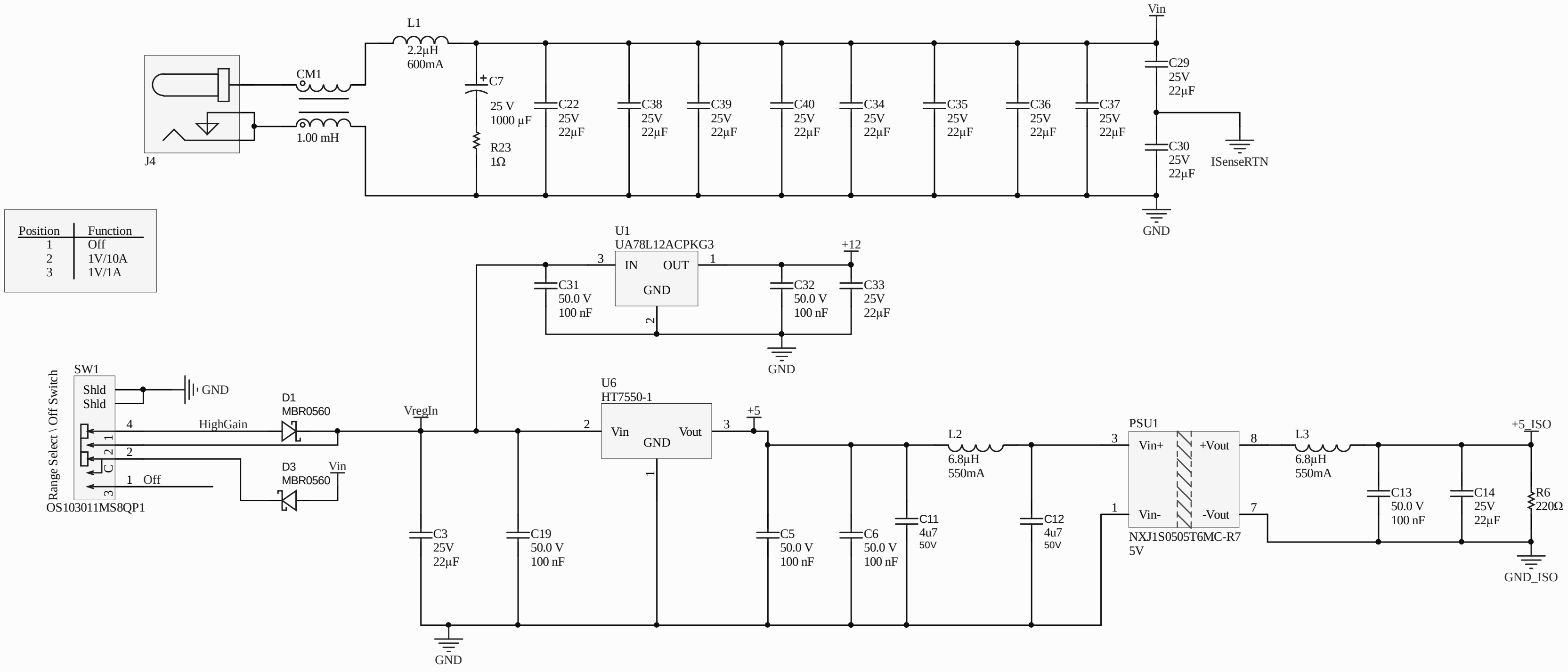
Power Supply


L1
2.2µH
600mA
Vin
CM1
1.00 mH
J4
C7
25 V
1000 µF
R23
1Ω
C22
25V
22µF
C38
25V
22µF
C39
25V
22µF
C40
25V
22µF
C34
25V
22µF
C35
25V
22µF
C36
25V
22µF
C37
25V
22µF
C29
25V
22µF
C30
25V
22µF
ISenseRTN
GND
U1
UA78L12ACPKG3
3
IN
OUT
1
GND
2
+12
C31
50.0 V
100 nF
C32
50.0 V
100 nF
C33
25V
22µF
GND
Range Select \ Off Switch
SW1
Shld
Shld
GND
4
HighGain
1
2
2
C
1 Off
3
OS103011MS8QP1
D1
MBR0560
D3
MBR0560
Vin
VregIn
U6
HT7550-1
2
Vin
GND
Vout
3
1
+5
C3
25V
22µF
C19
50.0 V
100 nF
GND
C5
50.0 V
100 nF
C6
50.0 V
100 nF
C11
4u7
50V
L2
6.8µH
550mA
C12
4u7
50V
PSU1
3
Vin+
+Vout
8
1
Vin-
-Vout
7
NXJ1S0505T6MC-R7
5V
L3
6.8µH
550mA
+5_ISO
C13
50.0 V
100 nF
C14
25V
22µF
R6
220Ω
GND_ISO

Position	Function
1	Off
2	1V/10A
3	1V/1A

Designed By
Shawn Melanson

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Technical Services\Shawn
Melanson Logo 4.png

Title		
Size B	Number	Revision
Date: 2026-08-20		Sheet of
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